

ML Assignment-2 (Group: 29)

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I. INTRODUCTION

In the task, three supervised learning classifiers i.e., naive Bayes (NB), Decision Tree (DT), and k-nearest neighbors (KNN) are used for comparison on a given spam base dataset.

To eliminate the improper consequence of dataset splitting, sufficient 10-fold cross-validation is used for dividing the dataset. Once the 10 times training is done on an individual fold then the computation and predictive performance are measured. Performance of computation is measured by calculating training time and predictive is measured by calculating accuracy and f-measure.

After the measurement process is complete then the sufficient difference between the three algorithms on each measured metric is sure to exist through the Friedman test. Hence while testing the critical difference is calculated between the three algorithms set by set by using the Nemenyi test. The experiment is done on 3 sets NB with DT, NB with KNN, and DT with KNN in terms of time complexity, accuracy, and f-measure.

The results show that according to the Nemenyi post-hoc test with training time and F-measure, there is no significant difference between (NB, DT) and (NB, KNN). But there is a significant difference between (DT, KNN). In accuracy, there is no significant difference between (NB, KNN). But there is a significant difference between (NB, DT) and (DT, KNN).

II. IMPLEMENTATION

A. Algorithm Selection:

I used Scikit-learn "sklearn" python library to import classifiers i.e. naive Bayes, decision tree, and k-nearest neighbors. However, these classifiers are commonly used by lots of researchers and they can do their experiments by using these classifiers and check their performance, therefore I chose them. The parameters are set manually and it's easy to initialize the classifier.

B. 10-fold cross-validation:

To implement the 10-fold cross-validation, firstly the given data set is stratified with class label spam and ham. Every subset has an equal proportion of spam/ham data which is (39.40 : 60.60). Then it's divided into the subset of 10-fold with equal size by the stratification thru the "StratifiedKFold" function import by "sklearn".

```
# 10-Fold cross-validation classification
splitTen = StratifiedKFold (shuffle = True, n_splits = 10)
# Stratified thru Y label value and the data is split into 10-fold
splitTen.split(xdata,ydata):
```

C. Training Model:

For implementing the training model, the "fit()" function is used, this function is provided by "sklearn" for each classifier. The divided train set is used for training models. For illustration:

```
# initializing model of Naive Bayes as GaussianNB
NBclf = GaussianNB()
# model training with training set
NBclf.fit(xtrain, ytrain)
```

D. Measurements:

- Training time consumption: First the start time is recorded. After that, the model is trained, and the end time is recorded. Then subtract the starting and ending time to calculate time.

```
# Time calculation of naive Bayes
startTime = time.time()
NBclf.fit(Xtrain, Ytrain)
endTime = time.time()
NBtime = endTime - startTime
```

- Accuracy: The function of "sklearn" "score()" is used to calculate the accuracy model on a testing set.

```
# Accuracy of naive bayes
NBacc = NBclf.score(Xtest, Ytest)
```

- F-measure: The function of "sklearn" "f1-score()" is used to calculate the f-measure on the testing set.

```
NBypred = NBclf.predict(Xtest)
NBf1 = f1_score(Ytest, NBypred)
```

E. Friedman Test:

The "scipy.stats" python library function "friedmanchisquare()" is used for calculating the Friedman test. Initially, the performance of each fold of each classifier is ranked, then the average rank ($\bar{r}$) of each classifier is calculated. Thus, according to the $\bar{r}$ the value of Q which is also known as the test statistic is calculated with the below equation:

$$Q = \frac{12n}{k(k+1)} \sum_{j=1}^k \left(\bar{r}_j - \frac{k+1}{2} \right)^2$$

Where k = no. of groups also known as treatments, n = no. of the subject, r_j = rank sum for the jth category.

Once Q value is known, then the value is compared with 7.82. In this case k = 3, n=10, and $\alpha = 0.05$ on the distribution of χ^2 . If the value of Q is higher, then the significant difference exists, else not.

F. Nemenyi Post-hoc Test:

If the Friedman test gives a positive result, then the Nemenyi test is executed. The Friedman test result value generated by average rank was used to give the input to Nemenyi Test. The difference between the set of average rank is compared with the CD (critical difference) calculated with the given formula.

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6n}}$$

As we have taken 3 algorithms to compare and we calculate the subset test results in 10 folds. Hence $n=10$, $k = 3$, and with $\alpha = 0.05$, the degree of freedom (df) = infinity (inf), which is 3.314, and $q_\alpha = \frac{3.314}{\sqrt{2}} = 2.343$ as given, then CD is calculated and we get $CD=1.0478$. if the average rank difference between the two algorithms is higher than the critical difference, then the result is a significant difference between the two classifiers, else not.

III. RESULTS AND OUTPUTS

A. *Performance calculation details*

The complete result for all measuring metrics of Steps 2,3 & 4 are present at the end of the code by the output displayed with the function call 'RunningAllSteps()', formatted in the form of Table 12.4 and Table 12.8.

B. *Friedman Test result*

For all of the training time consumption, accuracy and F-measure a significant difference is shown between the three classifiers.

C. *Nemenyi Post-hoc test result*

Nemenyi post-hoc test calculation proves that there is a significant difference between DT and KNN classifiers on all training time, accuracy, and F-measures. And there is no significant difference between NB and KNN on all training time, accuracy, and F-measure.

With NB and DT there is a significant difference in accuracy and no significant difference with training time and F-measure.